

Lesson 1 · Variables & Data Types

Module SC101 · Introduction to Python · Social Coding SA

What is a variable?

A variable is a named box in the computer's memory where we store a value. We choose the name; Python remembers what's inside. We create one with the equals sign — read it as "gets", not "equals":

```
name = "Amahle"
age = 16
height = 1.62
likes_coding = True
```

The four types you will use every day

str (string) — text, always in quotes: "Limpopo". **int** — whole numbers: 16, -3, 2025. **float** — decimal numbers: 1.62, 99.9. **bool** — True or False, nothing else.

Ask Python what type something is with `type()`:

```
print(type(age))      # <class 'int'>
print(type(name))     # <class 'str'>
```

Working with variables

```
marks = 72
marks = marks + 8      # marks is now 80
greeting = "Hello " + name
print(greeting)        # Hello Amahle
```

Rules for names: letters, numbers and underscores only; cannot start with a number; case matters (Age and age are different). Choose names that say what they hold: `total_marks` beats `tm`.

Common mistake

```
age = "16"             # this is TEXT, not a number!
age + 1                # TypeError - can't add int to str
int("16") + 1          # 17 - convert first
```

Try it now (5 min)

1. Create variables for your name, age, and school. 2. Print a sentence using all three. 3. Store your two favourite subjects and print them. 4. Predict the type of 3.0 — then check with `type()`.

Homework: read the cheat sheet and finish the four exercises above if not done in class.